

Data Cleaning

Part 2

```
library(tidyverse)
```

```

— Attaching core tidyverse packages ————— tidyverse 2.0.0
—
✓ dplyr      1.1.4      ✓ readr      2.1.5
✓ forcats    1.0.1      ✓ stringr    1.5.2
✓ ggplot2    4.0.0      ✓ tibble     3.3.0
✓ lubridate  1.9.4      ✓ tidyr      1.3.1
✓ purrr      1.1.0
— Conflicts ————— tidyverse_conflicts()
—
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

```

```
library(sf)
```

Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE

```
library(readr)
sv_raw <- "data/g24_sov_by_g24_svprec.csv"
sv_raw
```

```
[1] "data/g24_sov_by_g24_svprec.csv"
```

```
sv_preview <- read_lines(sv_raw, n_max = 50)
sv_preview
```

```
[1]
"COUNTY,FIPS,SVPREC,ADDIST,SVPREC_KEY,ELECTION,GEO_TYPE,CDDIST,SDDIST,BEDIST,TOTREG,DEMREG,REPREG
[2]
"1,06001,200100,14,06001200100,g24,svprec,12,7,2,3535,0,0,0,0,0,0,0,0,0,256,0,0,0,0,0,0,0,0,0,0
[3]
"1,06001,200100A,14,06001200100A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,2804,0,0,0,0,0,0,0,0,0,0
```

[illegible]

[28]
"1,06001,204900,14,06001204900,g24,svprec,12,7,2,1074,0,0,0,0,0,0,0,0,0,0,461,0,0,0,0,0,0,0,0,0,0,0
[29]
"1,06001,204900A,14,06001204900A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,388,0,0,0,0,0,0,0,0,0,0,0,
[30]
"1,06001,205200,14,06001205200,g24,svprec,12,7,2,4322,0,0,0,0,0,0,0,0,0,0,181,0,0,0,0,0,0,0,0,0,0,0,
[31]
"1,06001,205200A,14,06001205200A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,3525,0,0,0,0,0,0,0,0,0,0,0,
[32]
"1,06001,205700,14,06001205700,g24,svprec,12,7,2,3683,0,0,0,0,0,0,0,0,0,0,225,0,0,0,0,0,0,0,0,0,0,0,
[33]
"1,06001,205700A,14,06001205700A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,2966,0,0,0,0,0,0,0,0,0,0,0,
[34]
"1,06001,206200,14,06001206200,g24,svprec,12,7,2,2148,0,0,0,0,0,0,0,0,0,0,197,0,0,0,0,0,0,0,0,0,0,0,
[35]
"1,06001,206200A,14,06001206200A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,1589,0,0,0,0,0,0,0,0,0,0,0,
[36]
"1,06001,206900,14,06001206900,g24,svprec,12,7,2,1448,0,0,0,0,0,0,0,0,0,0,185,0,0,0,0,0,0,0,0,0,0,0,
[37]
"1,06001,206900A,14,06001206900A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,935,0,0,0,0,0,0,0,0,0,0,0,
[38]
"1,06001,207000,14,06001207000,g24,svprec,12,7,2,1983,0,0,0,0,0,0,0,0,0,0,135,0,0,0,0,0,0,0,0,0,0,0,
[39]
"1,06001,207000A,14,06001207000A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,1566,0,0,0,0,0,0,0,0,0,0,0,
[40]
"1,06001,207200,14,06001207200,g24,svprec,12,7,2,2522,0,0,0,0,0,0,0,0,0,0,208,0,0,0,0,0,0,0,0,0,0,0,
[41]
"1,06001,207200A,14,06001207200A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,1855,0,0,0,0,0,0,0,0,0,0,0,
[42]
"1,06001,207700,14,06001207700,g24,svprec,12,7,2,2802,0,0,0,0,0,0,0,0,0,0,804,0,0,0,0,0,0,0,0,0,0,0,
[43]
"1,06001,207700A,14,06001207700A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,1375,0,0,0,0,0,0,0,0,0,0,0,
[44]
"1,06001,208010,14,06001208010,g24,svprec,12,7,2,3594,0,0,0,0,0,0,0,0,0,0,275,0,0,0,0,0,0,0,0,0,0,0,
[45]
"1,06001,208010A,14,06001208010A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,2615,0,0,0,0,0,0,0,0,0,0,0,
[46]
"1,06001,208300,14,06001208300,g24,svprec,12,7,2,3642,0,0,0,0,0,0,0,0,0,0,345,0,0,0,0,0,0,0,0,0,0,0,
[47]
"1,06001,208300A,14,06001208300A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,2376,0,0,0,0,0,0,0,0,0,0,0,
[48]
"1,06001,208510,14,06001208510,g24,svprec,12,7,2,2566,0,0,0,0,0,0,0,0,0,0,331,0,0,0,0,0,0,0,0,0,0,0,
[49]
"1,06001,208510A,14,06001208510A,g24,svprec,12,7,2,0,0,0,0,0,0,0,0,0,0,0,1519,0,0,0,0,0,0,0,0,0,0,0,
[50]
"1,06001,208690,14,06001208690,g24,svprec,12,7,2,28,0,

```
sv <- read_csv(sv_raw, guess_max = 20000, col_types = cols(.default = "c"))

num_cols <- names(sv)[str_detect(names(sv), "(REG$|VOTE$|_Y$|_N$)")]
sv <- sv %>%
  mutate(across(all_of(num_cols), ~ str_replace_all(.x, ",", "")) |>
as.integer()))
```

Warning: There were 20 warnings in `mutate()`.
 The first warning was:
 i In argument: `across(...)`.
 Caused by warning:
 ! NAs introduced by coercion
 i Run `dplyr::last\_dplyr\_warnings()` to see the 19 remaining warnings.

```
glimpse(sv)
```

```
Rows: 51,123
Columns: 76
$ COUNTY      <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1",...
$ FIPS        <chr> "06001", "06001", "06001", "06001", "06001", "06001",
"0600...
$ SVPREC      <chr> "200100", "200100A", "200200", "200200A", "201400",
"201400...
$ ADDIST      <chr> "14", "14", "14", "14", "14", "14", "14", "14", "14",
"14",...
$ SVPREC_KEY  <chr> "06001200100", "06001200100A", "06001200200",
"06001200200A...
$ ELECTION    <chr> "g24", "g24", "g24", "g24", "g24", "g24", "g24", "g24",
"g2...
$ GEO_TYPE    <chr> "svprec", "svprec", "svprec", "svprec", "svprec",
"svprec",...
$ CDDIST      <chr> "12", "12", "12", "12", "12", "12", "12", "12", "12",
"12",...
$ SDDIST      <chr> "7", "7", "7", "7", "7", "7", "7", "7", "7", "7",
"7",...
$ BEDIST      <chr> "2", "2", "2", "2", "2", "2", "2", "2", "2", "2",
"2",...
$ TOTREG      <int> 3535, 0, 2442, 0, 3773, 0, 541, 0, 1105, 0, 948, 0, 2721,
0...
$ DEMREG      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,0...
$ REPREC      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,0...
$ AIPREG      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,0,
```

```

0,...
$ GRNREG      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ LIBREG      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ NLPREG      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ REFREG      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ DCLREG      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ MSCREG      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ TOTVOTE     <int> 256, 2804, 262, 1816, 283, 2782, 89, 343, 394, 297, 837,
29...
$ DEMVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ REPVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ AIPVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ GRNVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ LIBVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ NLPVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ REFVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ DCLVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ MSCVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ PRCVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ ABSVOTE     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,...
$ ASSAIP01    <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",...
$ ASSDEM01    <chr> "94", "444", "117", "348", "107", "588", "45", "105",
"181"...
$ ASSDEM02    <chr> "110", "2023", "91", "1243", "128", "1841", "24", "172",
"1...
$ ASSREP01    <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",...
$ ASSREP02    <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",...
$ CNGDEM01    <chr> "102", "1668", "108", "1063", "139", "1688", "35", "192",

```

```

" ...
$ CNGDEM02 <chr> "102", "771", "99", "513", "98", "739", "38", "93", "143",
...
$ CNGIND01 <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",...
$ CNGREP01 <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",...
$ CNGREP02 <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",...
$ PRSAIP01 <chr> "3", "10", "1", "6", "3", "13", "2", "2", "3", "4", "5",
"2...
$ PRSD01 <chr> "181", "2562", "207", "1647", "231", "2522", "73", "297",
" ...
$ PRSGRN01 <chr> "9", "48", "13", "41", "9", "52", "0", "13", "6", "13",
"23...
$ PRSLIB01 <chr> "1", "10", "2", "3", "4", "13", "0", "0", "3", "2", "3",
"0...
$ PRSPAF01 <chr> "5", "17", "7", "12", "8", "32", "2", "4", "3", "7", "16",
...
$ PRSREP01 <chr> "51", "108", "26", "83", "17", "111", "11", "23", "55",
"24...
$ PR_2_N <int> 58, 493, 45, 342, 39, 399, 17, 45, 52, 38, 75, 28, 38,
344,...
$ PR_2_Y <int> 169, 2156, 196, 1385, 226, 2231, 66, 278, 309, 230, 706,
23...
$ PR_32_N <int> 78, 636, 55, 439, 74, 536, 14, 60, 81, 67, 152, 52, 39,
497...
$ PR_32_Y <int> 148, 1966, 187, 1261, 190, 2070, 68, 255, 274, 194, 607,
20...
$ PR_33_N <int> 136, 1774, 105, 1092, 124, 1509, 30, 127, 116, 96, 254,
96,...
$ PR_33_Y <int> 86, 784, 126, 584, 133, 1053, 49, 177, 231, 158, 471, 159,
...
$ PR_34_N <int> 123, 1485, 121, 1027, 144, 1515, 46, 186, 166, 141, 332,
12...
$ PR_34_Y <int> 98, 980, 105, 601, 96, 941, 33, 107, 174, 106, 398, 115,
36...
$ PR_35_N <int> 54, 581, 45, 419, 58, 563, 20, 57, 61, 50, 90, 28, 49,
551,...
$ PR_35_Y <int> 171, 2003, 188, 1261, 196, 1988, 58, 248, 293, 209, 677,
22...
$ PR_36_N <int> 106, 1356, 142, 888, 146, 1487, 49, 197, 214, 167, 433,
158...
$ PR_36_Y <int> 118, 1223, 99, 786, 119, 1084, 31, 112, 144, 95, 330, 96,
4...
$ PR_3_N <int> 51, 133, 25, 116, 33, 152, 10, 26, 38, 20, 72, 24, 12, 99,
...
$ PR_3_Y <int> 183, 2553, 220, 1646, 240, 2508, 74, 295, 340, 250, 718,

```

```

24...
$ PR_4_N    <int> 52, 381, 37, 271, 37, 330, 14, 41, 40, 24, 64, 25, 35,
316,...
$ PR_4_Y    <int> 181, 2294, 209, 1472, 231, 2316, 68, 279, 331, 241, 722,
24...
$ PR_5_N    <int> 94, 961, 66, 605, 63, 742, 19, 76, 72, 62, 133, 45, 45,
639...
$ PR_5_Y    <int> 132, 1660, 168, 1096, 197, 1862, 61, 240, 295, 199, 633,
21...
$ PR_6_N    <int> 75, 607, 59, 407, 57, 532, 17, 53, 85, 58, 189, 48, 36,
455...
$ PR_6_Y    <int> 143, 1958, 180, 1274, 196, 2029, 62, 257, 266, 200, 550,
20...
$ SENDEM01  <chr> "107", "1719", "101", "1102", "136", "1578", "37", "153",
"..."
$ SENDEM02  <chr> "103", "809", "114", "516", "105", "908", "34", "133",
"174..."
$ SENREP01  <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",..."
$ SENREP02  <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",..."
$ USPDEM01  <chr> "172", "2461", "199", "1572", "217", "2444", "67", "285",
"..."
$ USPREP01  <chr> "53", "155", "34", "111", "32", "153", "11", "23", "53",
"3..."
$ USSDEM01  <chr> "173", "2487", "207", "1593", "222", "2478", "67", "288",
"..."
$ USSREP01  <chr> "55", "155", "29", "109", "33", "151", "12", "23", "51",
"2..."

```

```

sv %>%
  summarise(across(all_of(num_cols), ~ sum(is.na(.x)))))

```

```

# A tibble: 1 × 42
  TOTREG DEMREG REPREG AIPREG GRNREG LIBREG NLPREG REFREG DCLREG MSCREG
TOTVOTE
  <int> <int> <int> <int> <int> <int> <int> <int> <int> <int>
1      0      0      0      0      0      0      0      0      0      0
0
# i 31 more variables: DEMVOTE <int>, REPVOTE <int>, AIPVOTE <int>,
#   GRNVOTE <int>, LIBVOTE <int>, NLPVOTE <int>, REPVOTE <int>, DCLVOTE <int>,
#   MSCVOTE <int>, PRCVOTE <int>, ABSVOTE <int>, PR_2_N <int>, PR_2_Y <int>,
#   PR_32_N <int>, PR_32_Y <int>, PR_33_N <int>, PR_33_Y <int>, PR_34_N <int>,
#   PR_34_Y <int>, PR_35_N <int>, PR_35_Y <int>, PR_36_N <int>, PR_36_Y <int>,

```

```
# PR_3_N <int>, PR_3_Y <int>, PR_4_N <int>, PR_4_Y <int>, PR_5_N <int>,  
# PR_5_Y <int>, PR_6_N <int>, PR_6_Y <int>
```

```
dir.create("data/clean", showWarnings = FALSE)  
write_csv(sv, "data/clean/g24_svprec_clean.csv")
```

Part 5

```
library(sf)  
library(dplyr)  
library(readr)  
  
dir.create("data/derived", recursive = TRUE)
```

```
Warning in dir.create("data/derived", recursive = TRUE): 'data/derived'  
already  
exists
```

```
sr_votes <- read_csv("/Users/estebanlopez/Downloads/  
state_g24_sov_data_by_g24_srprec 2.csv")
```

```
Rows: 25245 Columns: 76  
— Column specification
```

```
Delimiter: ","  
chr (49): FIPS, SRPREC, ELECTION, SRPREC_KEY, GEO_TYPE, ASSAIP01,  
ASSDEM01, ...  
dbl (27): COUNTY, ADDIST, CDDIST, SDDIST, BEDIST, TOTREG, DEMREG, REPREG,  
AI...
```

```
i Use `spec()` to retrieve the full column specification for this data.  
i Specify the column types or set `show_col_types = FALSE` to quiet this  
message.
```

```
sr_shp <- st_read("/Users/estebanlopez/Downloads/srprec_state_g24_v01_shp/  
srprec_state_g24_v01_shp.shp")
```

```
Reading layer `srprec_state_g24_v01_shp' from data source  
`/Users/estebanlopez/Downloads/srprec_state_g24_v01_shp/  
srprec_state_g24_v01_shp.shp'  
using driver `ESRI Shapefile'
```



```
Warning in CPL_read_ogr(dsn, layer, query, as.character(options), quiet, :
GDAL
Message 1:
/Users/estebanlopez/Downloads/srprec_state_g24_v01_shp/
srprec_state_g24_v01_shp.shp
contains polygon(s) with rings with invalid winding order. Autocorrecting
them,
but that shapefile should be corrected using ogr2ogr for example.
```

```
Simple feature collection with 24224 features and 6 fields
Geometry type: MULTIPOLYGON
Dimension: XY
Bounding box: xmin: -124.482 ymin: 32.52883 xmax: -114.1312 ymax: 42.0095
Geodetic CRS: NAD83
```

```
ab604 <- st_read("/Users/estebanlopez/Downloads/AB604/AB604.shp")
```

```
Reading layer `AB604' from data source
`/Users/estebanlopez/Downloads/AB604/AB604.shp' using driver `ESRI
Shapefile'
Simple feature collection with 52 features and 15 fields
Geometry type: MULTIPOLYGON
Dimension: XY
Bounding box: xmin: -13857270 ymin: 3832931 xmax: -12705030 ymax: 5162404
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
sr_shp <- sr_shp %>% st_transform(3310) %>% st_make_valid()
ab604 <- ab604 %>% st_transform(3310) %>% st_make_valid()
sr_shp <- sr_shp %>%
  left_join(sr_votes, by = "SRPREC_KEY") %>%
  mutate(precinct_dem_votes = as.numeric(CNGDEM01) + as.numeric(CNGDEM02),
  precinct_rep_votes = as.numeric(CNGREP01) + as.numeric(CNGREP02)) %>%
  replace_na(list(precinct_dem_votes = 0, precinct_rep_votes = 0))
```

```
Warning: There were 4 warnings in `stopifnot()`.
The first warning was:
i In argument: `precinct_dem_votes = as.numeric(CNGDEM01) +
  as.numeric(CNGDEM02)`.
Caused by warning:
! NAs introduced by coercion
i Run `dplyr::last_dplyr_warnings()` to see the 3 remaining warnings.
```

```
intersections <- st_intersection(sr_shp, ab604 %>% select(ab604_district =
DISTRICT))
```

Warning: attribute variables are assumed to be spatially constant throughout all geometries

```
intersections <- intersections %>%
  mutate(area_piece = st_area(geometry)) %>%
  group_by(SRPREC_KEY) %>%
  mutate(total_sr_area = sum(area_piece),
         frac = as.numeric(area_piece / total_sr_area)) %>%
  ungroup()
intersections <- intersections %>%
  mutate(DEMVOTE_alloc = precinct_dem_votes * frac, REPVOTE_alloc =
precinct_rep_votes * frac)
ab604_est <- intersections %>%
  st_set_geometry(NULL) %>%
  group_by(ab604_district) %>%
  summarise(DEMVOTE = sum(DEMVOTE_alloc, na.rm = TRUE),
           REPVOTE = sum(REPVOTE_alloc, na.rm = TRUE))
write_csv(ab604_est, "data/derived/ab604_estimated_votes_area_weighted.csv")
print(ab604_est)
```

```
# A tibble: 52 × 3
  ab604_district DEMVOTE REPVOTE
  <chr>          <dbl>   <dbl>
1 01            182133. 150292.
2 02            231720. 148291.
3 03            190157. 164795.
4 04            190078. 144246.
5 05            117757. 226153.
6 06            170485. 151682.
7 07            188945. 143806.
8 08            206370.  96002.
9 09            151255. 107582.
10 10           241204. 122128.
# i 42 more rows
```